

Interim presentation Drone VSLAM

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Bitcraze



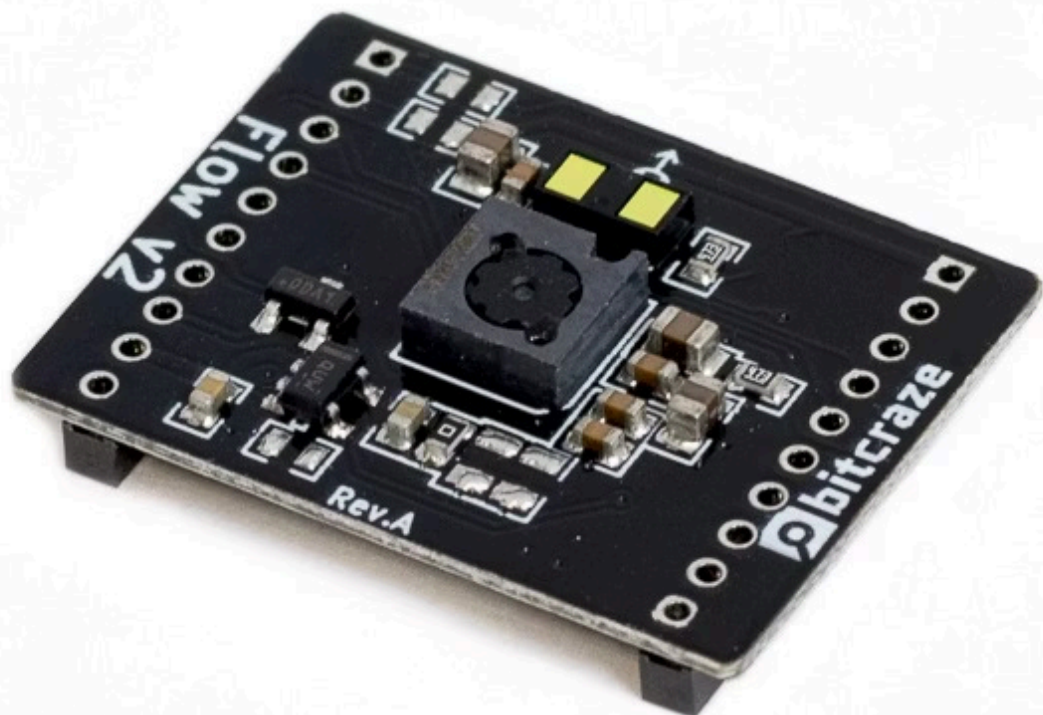
Crazyflie 2.1+



Crazyradio 2.0

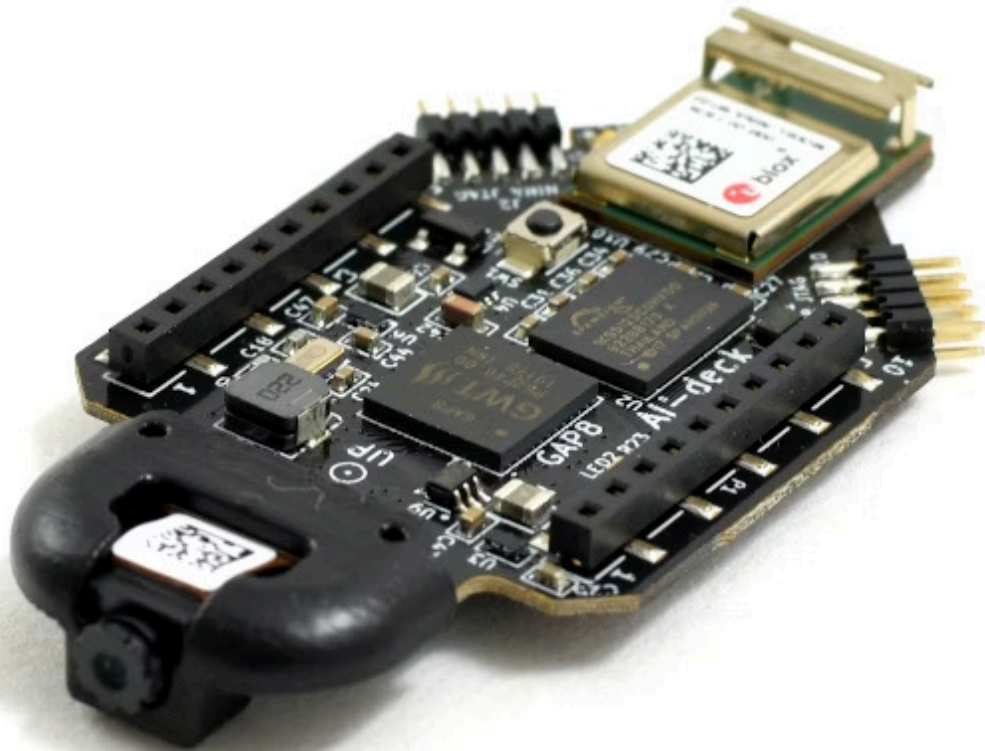


Crazyflie decks



Flow deck v2

AI deck 1.1

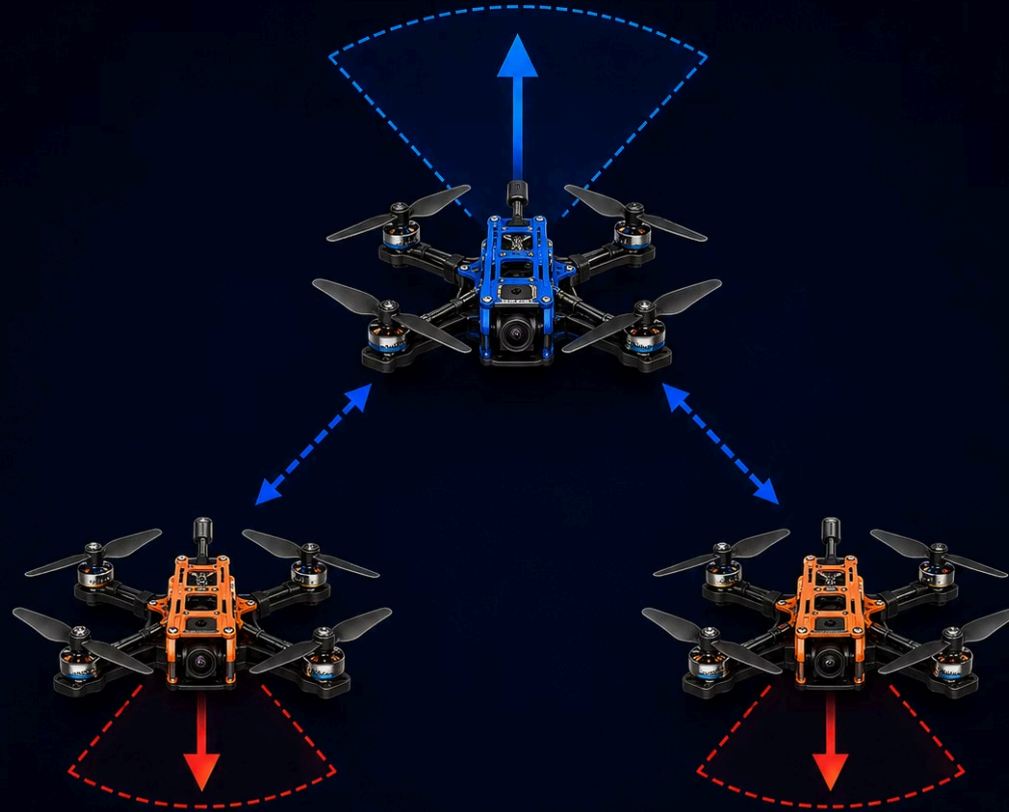




Multi-Ranger Deck

Leader drone

Multi-ranger deck + Flow deck



Analyzing drone 1

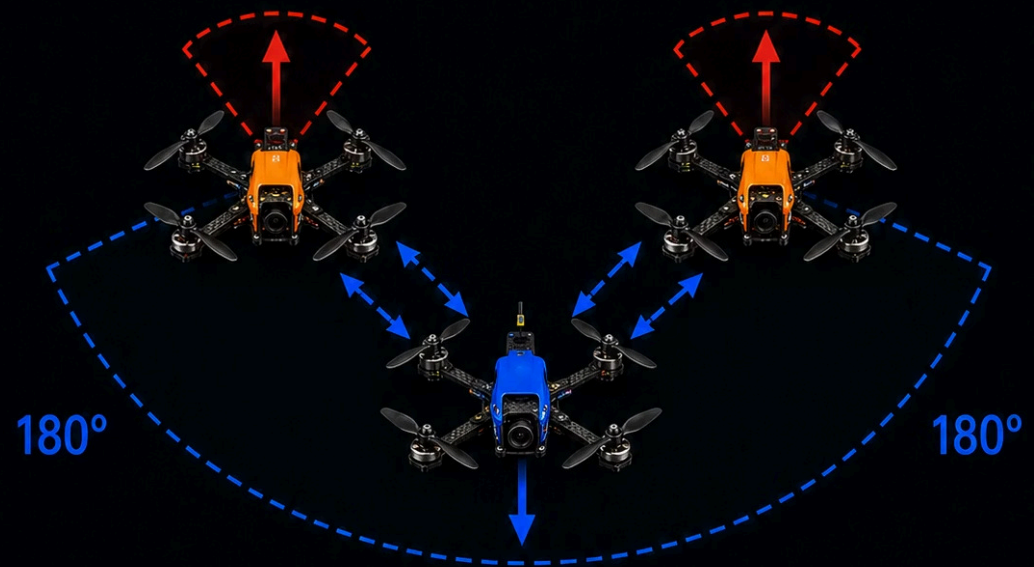
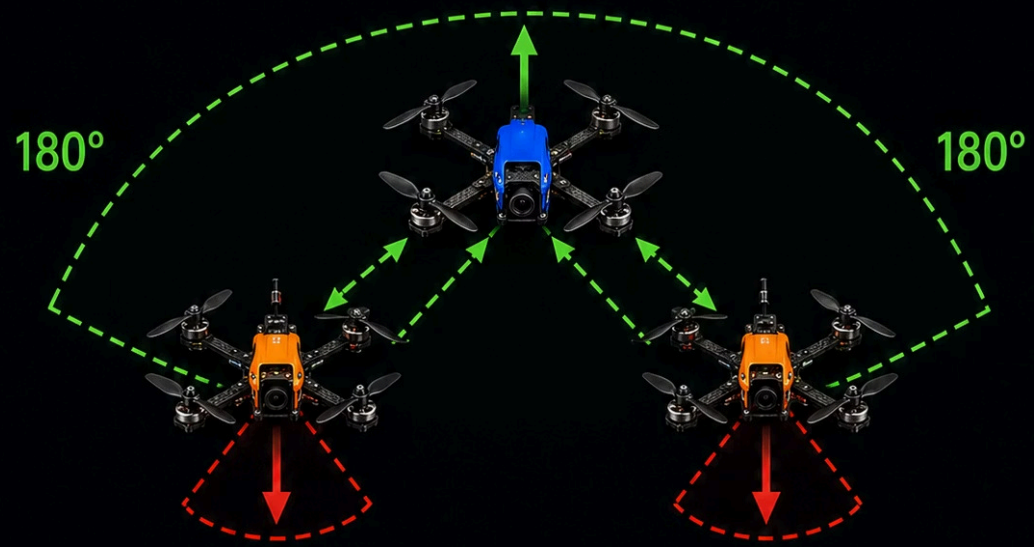
AI deck + Flow deck

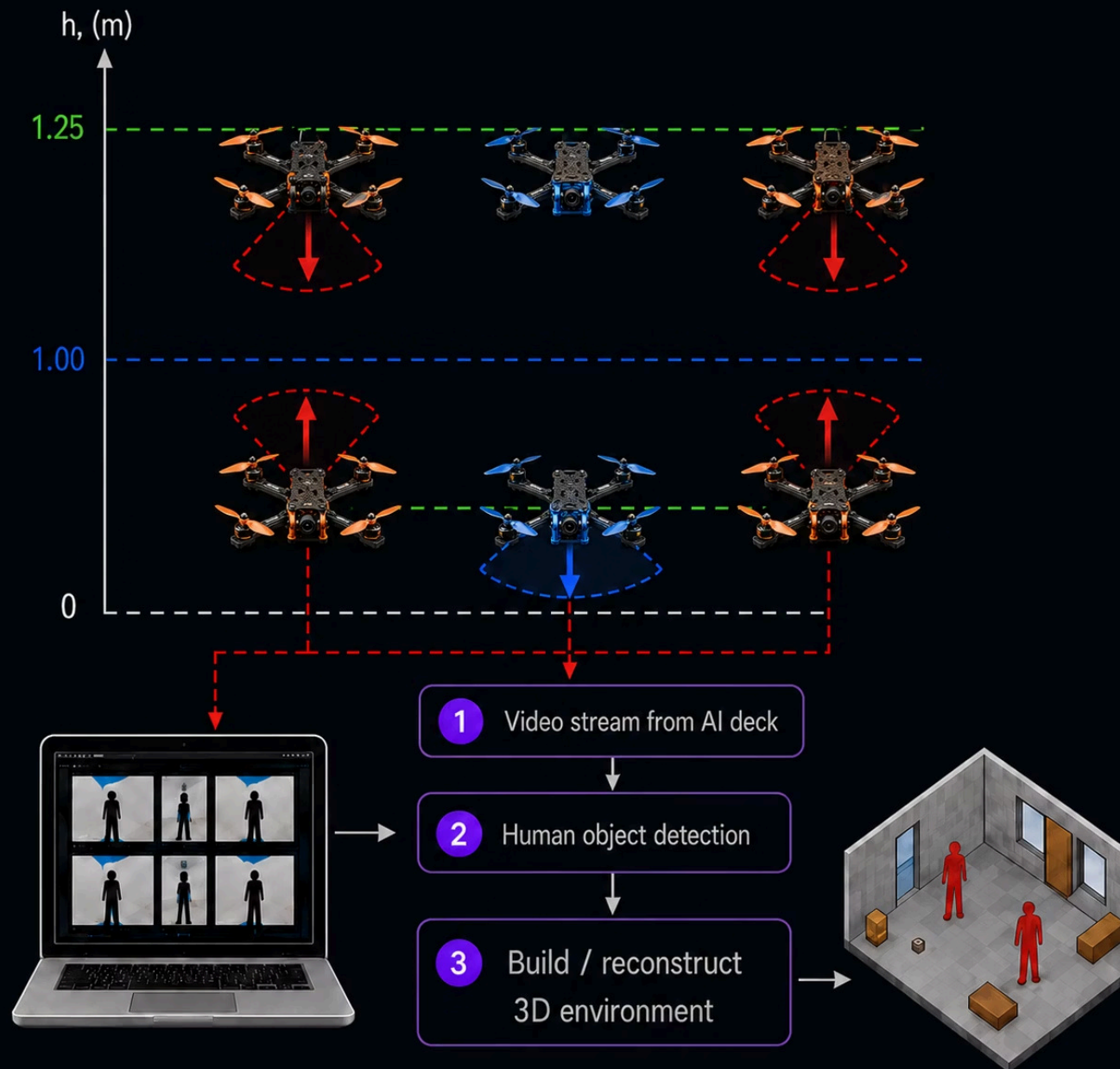
Human object detection
+ video stream

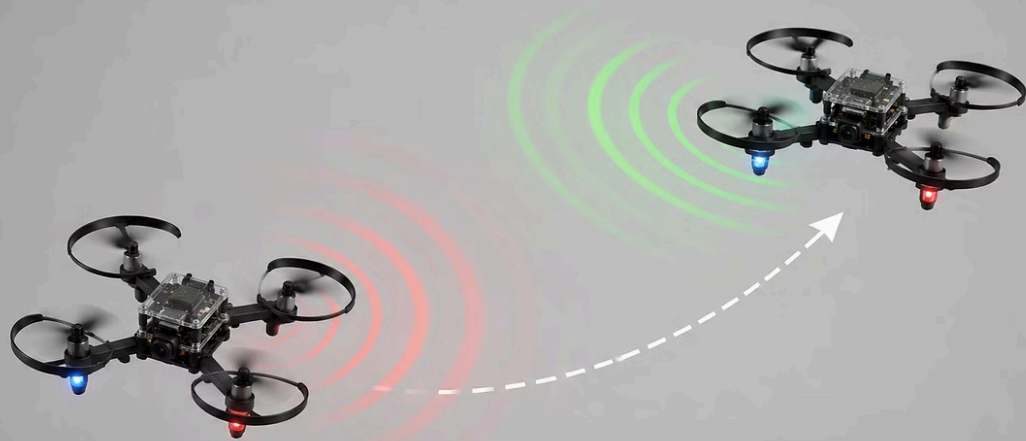
Analyzing drone 2

AI deck + Flow deck

Human object detection
+ video stream

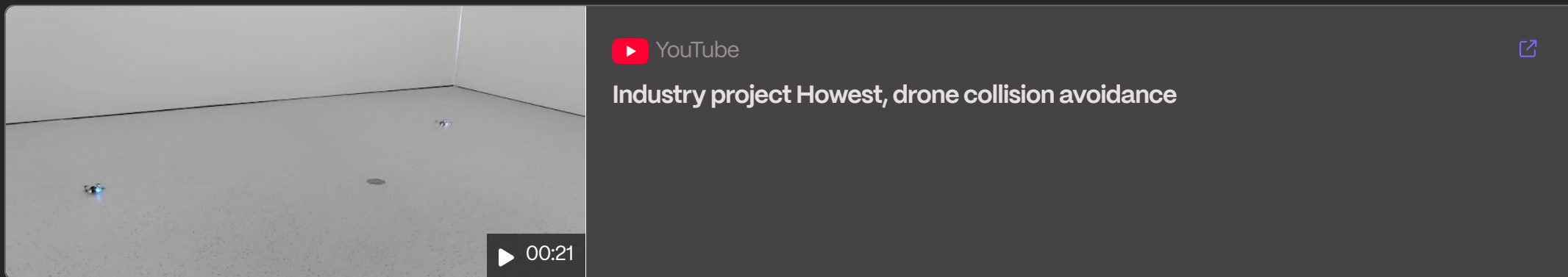


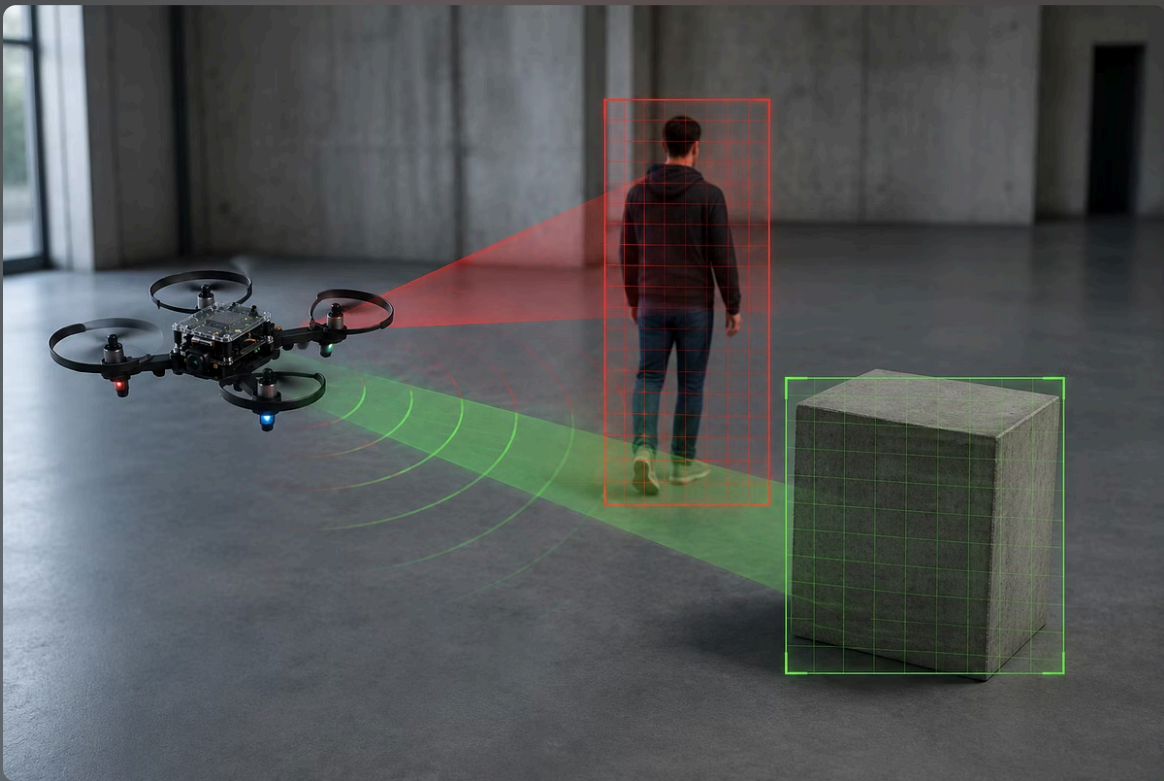




Drone Avoidance

Collision avoidance demo

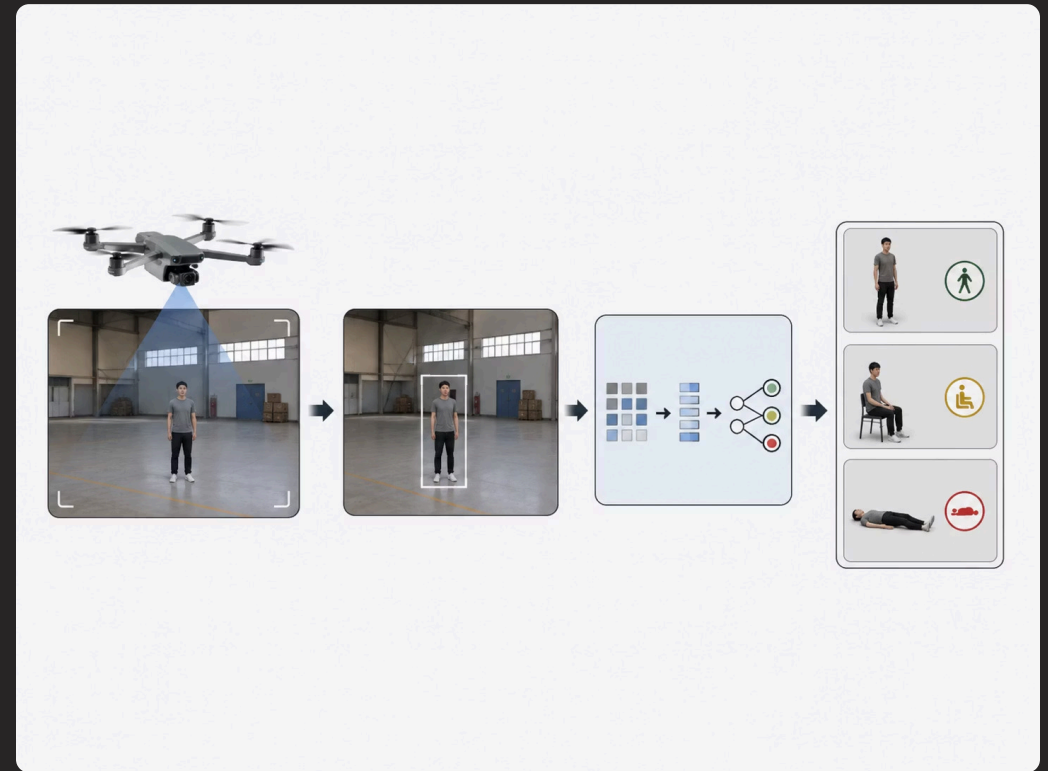




Object Detection

Our Objective

- Locate people
- Classify physical state
- Calculate people amount



First stage: Human detection

- Drone images → fine tuned YOLOv26n
- Detection of the humans

Second Stage: Human Posture Classification

- Cropped bounding boxes → second model
- Classification of the cropped bounding box
- Number of people estimation

Why object detection and posture classification are separated?

- Better robustness
- Independent failure condition
- Higher training set for human detector
- Conditional compute

Dataset Used for Human Detection

Dataset size

920 images

1,635 labeled person instances

Source

Drone-captured images

Annotation format

Bounding boxes

Class: person

Retraining results

Precision

0.78 → 0.84

+8% improvement

Recall

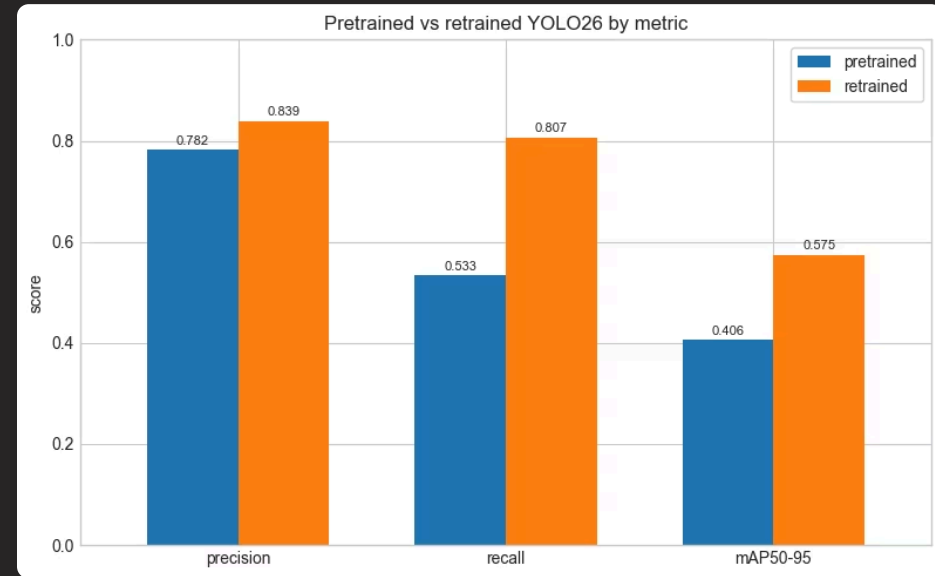
0.53 → 0.81

+28% improvement

mAP50-95

0.40 → 0.58

+18% improvement



Dataset we will use for human classification

Dataset size

Approximately 1200 images
400 images per class

Source

Drone-captured images

Annotation format

Bounding boxes

Classes:

- person_standing
- person_sitting
- person_laying

Future pipeline

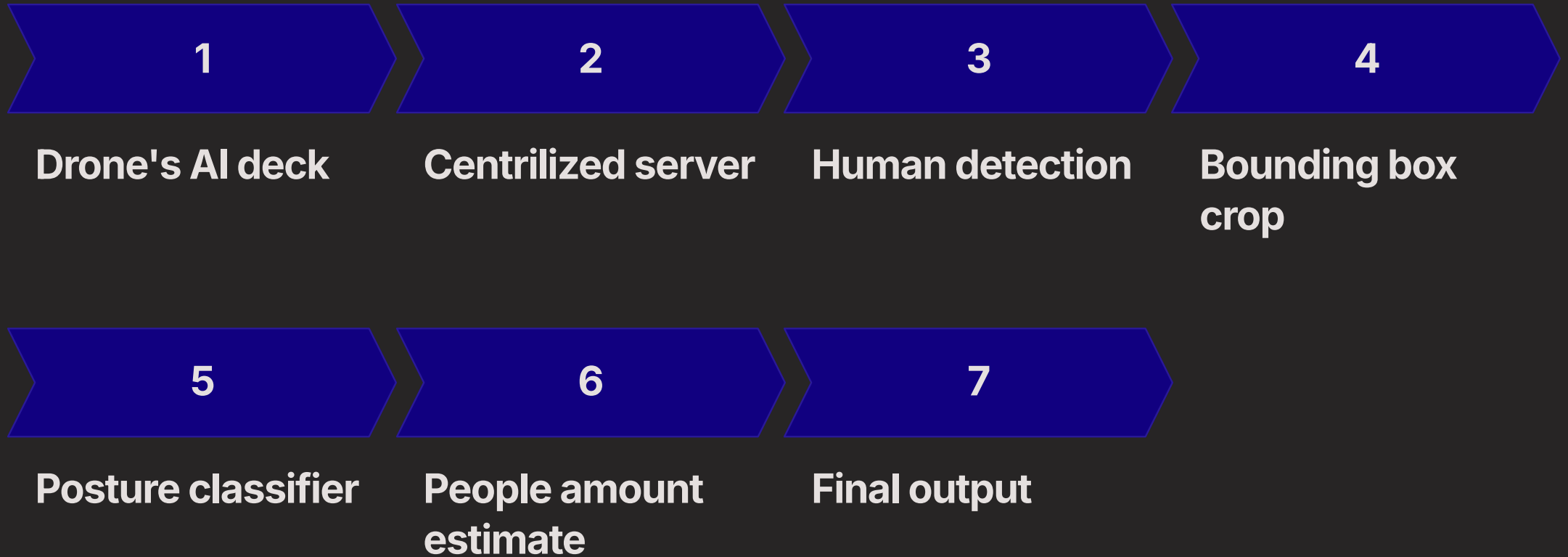


Image quality constraints

- Low resolution images
- Motion blur
- Light blur
- Bad feature embeddings







Filtering logic



Human Detection

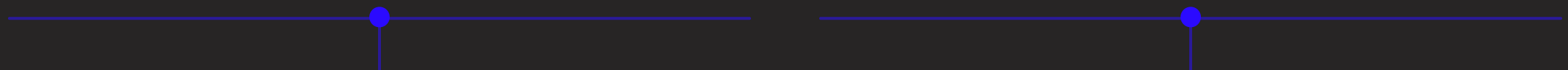
- High error cost
- Low confidence threshold



Posture Classification

- Higher confidence threshold
- Low confidence → person_unknown

Inference time



Current

27ms

One image needs 27 ms to be inferenced.

With both models

35–45ms

With both models it will take approximately 35–45 ms, which is still acceptable for us.

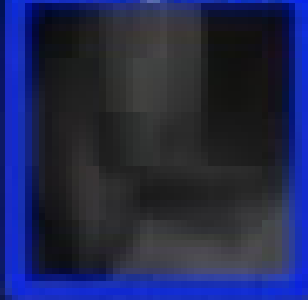


But model will be quantized if possible to further reduce the time needed for inference

Detection Examples



Human 0.88



Human 0.96



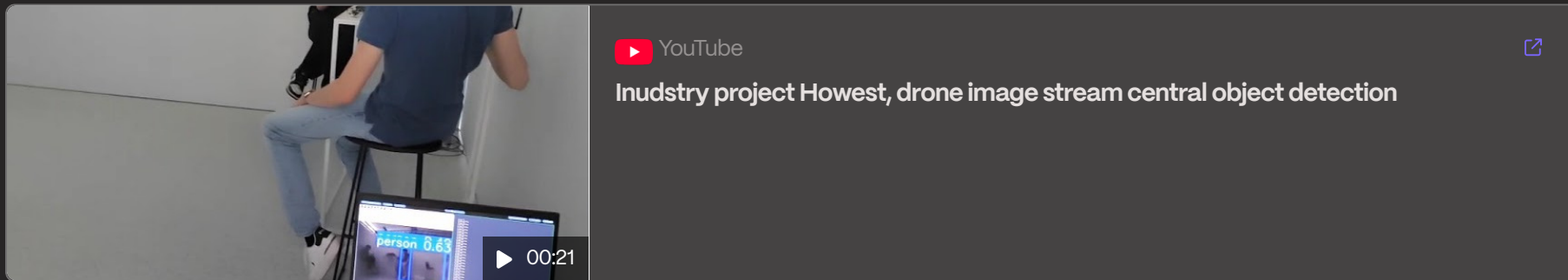
Human 0.96





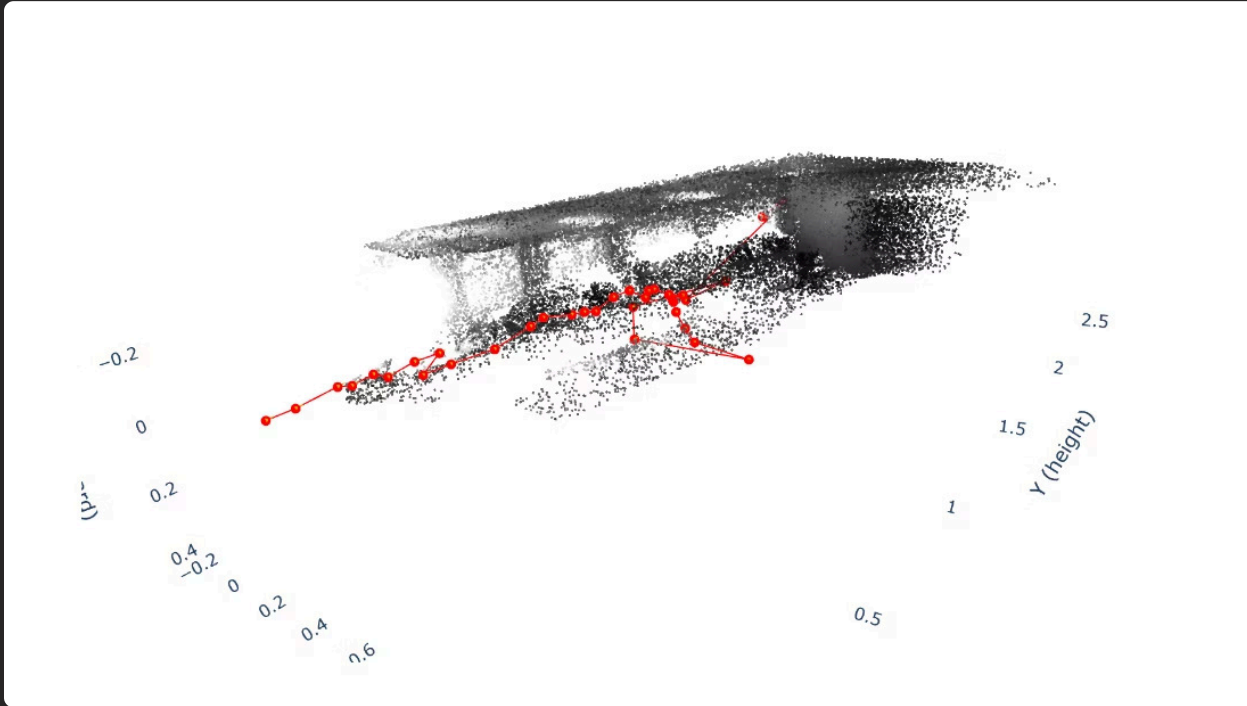
Human	0.95
Human	0.78

Drone image stream object detection on central compute



Mapping + Localisation

Mapping and localisation support navigation and positioning.



- Pointcloud for navigation
- Absolute position reference

Pipeline

01

Frame Pre-processing

Prepare each incoming frame for mapping.



Image-to-Points (I2P)

Convert the image into local 3D points.



Local-to-World (L2W)

Merge local points into the world pointcloud.

The Mapping API processes one frame at a time, adding new points to an existing pointcloud and returning the camera position.

Limitations



No depth sensors

Depth is estimated with a model.



Outside the map

Position estimates become inaccurate.

Management



To manage it all

☐	Open	45	Closed	59	Author ▾	Labels ▾	Projects ▾
☐	Retrospective week 5	management	1 Effort Low	#108 · roel-remmerie opened yesterday			
☐	Retrospective week 4	management	1 Effort Low	#107 · roel-remmerie opened yesterday			
☐	Retrospective week 3	management	1 Effort Low	#106 · roel-remmerie opened yesterday			
☐	Create a ranger deck example	code	research	testing	2 Effort Medium	#105 · roel-remmerie opened yesterday	
☐	Preparing live demo for interim presetaion (video)	code	management	2 Effort Medium	#104 · roel-remmerie opened yesterday		
☐	Presenting interim presentation	management	1 Effort Low	#103 · roel-remmerie opened yesterday			
☐	Retrospective week 2	management	1 Effort Low	#102 · roel-remmerie opened yesterday			
☐	Find a way to calculate amount of people on sequential images in the room	research	1 Effort Low	#100 · NazarMikhin opened yesterday			
☐	Brainstorm system architecture	research	1 Effort Low	#99 · roel-remmerie opened yesterday			
☐	Make model classify person's state (sitting, laying, standing)	code	3 Effort High	#98 · NazarMikhin opened yesterday			
☐	Getting IRL obstacles	material	1 Effort Low	#97 · roel-remmerie opened yesterday			

I got 99 issues but a bug ain't one

Scrum + New view

Filter by keyword or by field

View

Backlog 27

Entire project backlog

- vslam #76
Make a pointcloud map from the image stream of a single route flying drone on central compute
3 Effort High
- vslam #79
Create a visual interface for tracking drone taken path and current position
3 Effort High
- vslam #98
Make model classify person's state (sitting, laying, standing)
3 Effort High
- vslam #77
Infer the position of a single flying drone using its image stream and a pointcloud map on central compute
3 Effort High
- vslam #75
Make a pointcloud map from the 360 view image stream of a single flying drone on central compute
2 Effort Medium
- vslam #105

Todo 6

This item hasn't been started

- vslam #104
Preparing live demo for interim presentation (video)
2 Effort Medium
- vslam #96
Dockerize mapping api
1 Effort Low
- vslam #97
Getting IRL obstacles
1 Effort Low
- vslam #99
Brainstorm system architecture
1 Effort Low
- vslam #103
Presenting interim presentation
1 Effort Low
- vslam #94
Literature study on swarm navigation
1 Effort Low

In progress 6

This is actively being worked on

- vslam #55
Fork SLAM3R to have api for updating map, getting abs coordinates, getting map.
5 Effort Very High
- vslam #89
Make a interim presentation
2 Effort Medium
- vslam #86
Make a static IP for each drone that matches the drone ID
2 Effort Medium
- vslam #93
Adding obstacles avoidance system to AI deck
2 Effort Medium
- vslam #64
Find the most suitable model for object detection
1 Effort Low
- vslam #102
Retrospective week 2
1 Effort Low

Done 58

This has been completed

- vslam #74
Perform object detection on the image stream from a non flying drone using YOLO26 nano on central compute
3 Effort High
- vslam #39
Test creating a 3D environment with the recorded video in class
3 Effort High
- vslam #37
Fix the AI deck vision crashing problem by changing resolution and FPS
3 Effort High
- vslam #22
Learn C
3 Effort High
- vslam #71
Adapt and improve the found repositories
3 Effort High
- vslam #30
Determine if we will be able to retrieve coordinates to use honeybee navigation
2 Effort Medium

abandoned 3

This no longer belongs in the backlog

- vslam #54
Every week meeting with client with reporting about all researches and testing
3 Effort High
- vslam #52
Research the functionality of all the decks and choose 2 the most useful
2 Effort Medium
- vslam #69
Create basic drone flight interface
2 Effort Medium

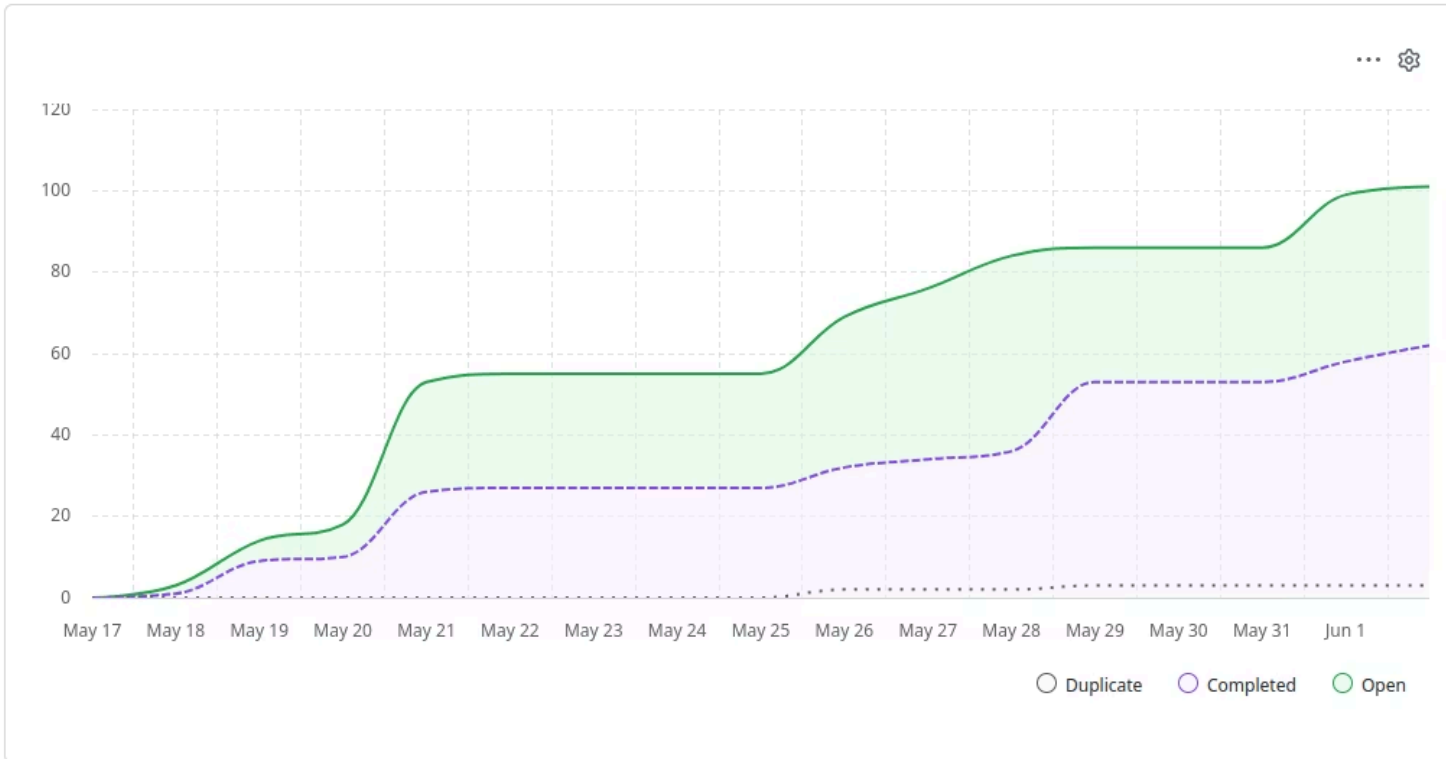
Burn up

 Configure

🔍 Filter by keyword or by field

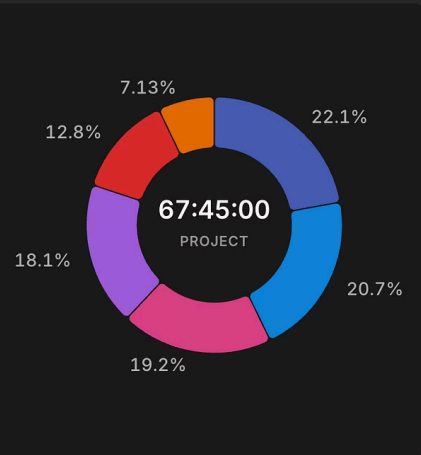
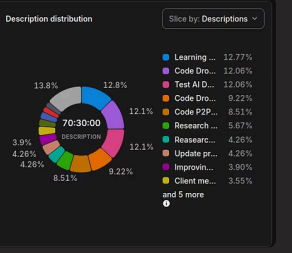
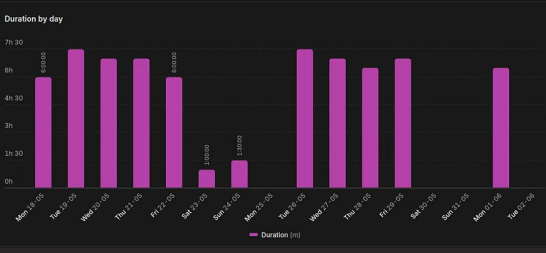
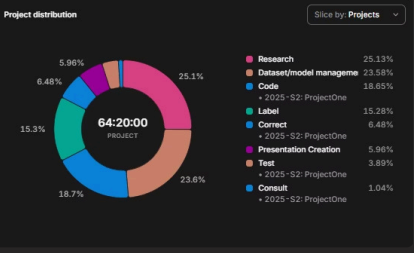
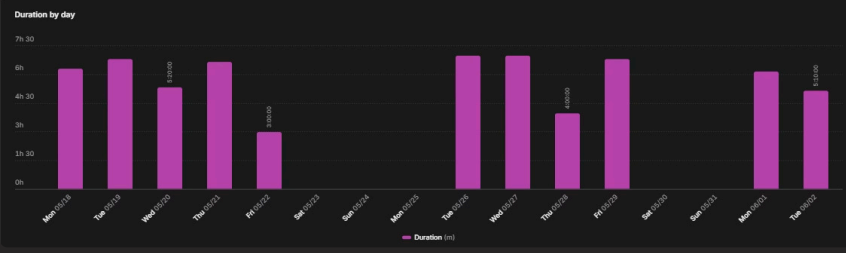
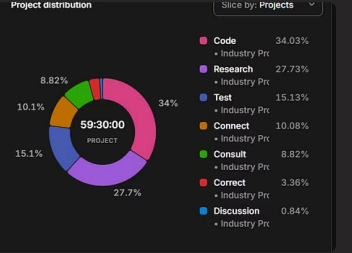
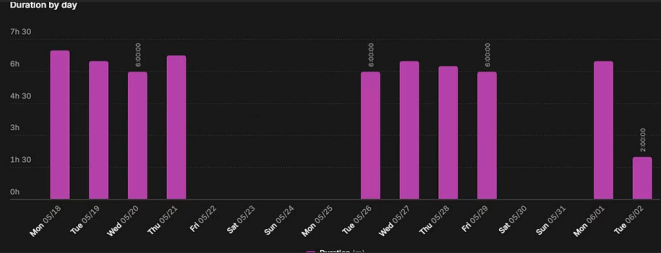
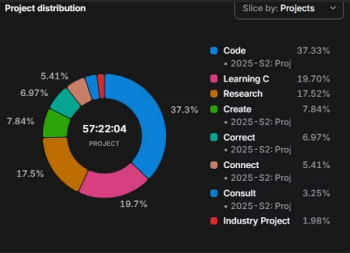
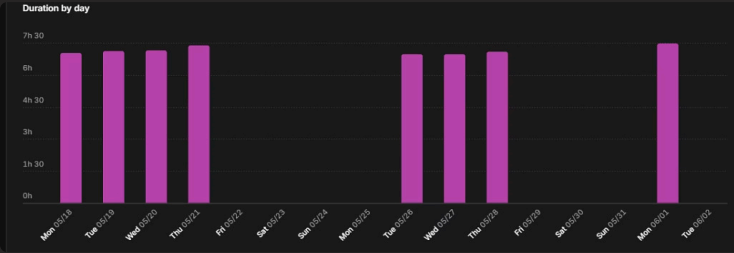
2 weeks 1 month 3 months Max

📅 May 17 - Jun 2 ▾





Toggl





Retrospective